



BATTERYLESS MICROCONTROLLER Operation Using Supercapacitor

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Imagine a world where microcontrollers (MCUs) and development boards operate without batteries. Energy-harvesting technology now makes this a practical reality. Several batteryless IoT sensors already capture power from ambient sources such as light, heat, vibration, and RF signals, enabling fully autonomous operation. Engineered for ultra-low-power performance, these devices typically consume only a few microamps to milliamps and employ multiple deep-sleep modes to maximise energy efficiency.

For prototyping, learning, or experimental work, engineers often need a general-purpose microcontroller rather than a specialised sensor. This design introduces a proof-of-concept system that

demonstrates continuous operation of an MCU paired with an OLED display, entirely without a battery and without relying on sleep modes. Ambient energy is harvested, stored in a supercapacitor, and used to power both the MCU and the display continuously.

The system intentionally showcases a comparatively power-hungry configuration, including an always-active OLED display, yet still operates without a battery. During testing, the complete setup, comprising the MCU and OLED display with active processing, ran for up to five minutes on a single charge in normal operating mode.

When configured for ultra-low-power operation, the system continued to function for up to 25 minutes

without an external power source, drawing only 6-7mA in total. Additional optimisation remains possible through more energy-efficient components, reduced display refresh rates, lower brightness levels, or improved duty-cycling techniques.

Fig. 1 shows the author's prototype of an MCU and OLED display running entirely without a battery.

Bill of materials

During the testing phase, the LTC3588 energy-harvesting IC was used along with a supercapacitor to collect and store ambient energy. However, the supercapacitor's charging time was relatively long. To overcome this limitation, two specialised energy-harvesting ICs were incorporated into an improved

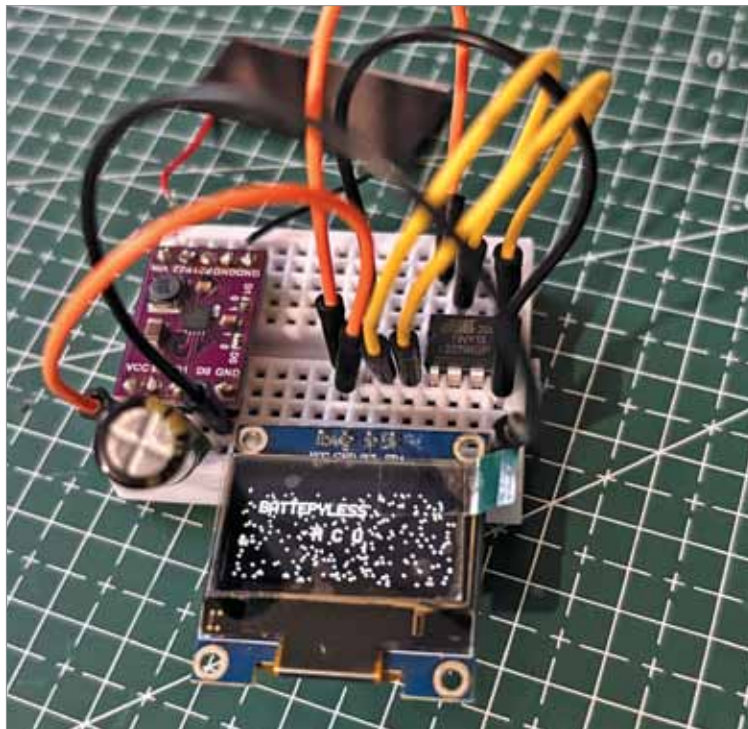


Fig. 1: Author's prototype

TABLE 1
BILL OF MATERIALS

Components	Designator	Quantity
Piezo transducer	PZ1, PZ2	2
Supercapacitor 2.7V, 6F	C1, C2	2
10 μ F capacitor	C3, C7	2
10nF capacitor	C4	1
1 μ F capacitor	C5	1
47 μ F capacitor	C6	1
1N5819 diode	D1, D2	2
Inductor coil (6SWG copper enamelled wire around 60 turns air core)	L1	1
10 μ H inductor coil	L2	1
SSD1306 OLED display	LCD1	1
Push switch (normally open)	SW1	1
Solar cell 3V, 50mA	PV1	1
9.9k Ω resistor	R1	1
2.26M Ω resistor	R2	1
1M Ω resistor	R3	1
LTC3588 energy harvesting IC	U1	1
LTC3105EMS energy harvesting IC	U2	1
ATtiny13A microcontroller	U3	1